CUSTOMER BEHAVIOR ANALYSIS

ALFIDO TECH INTERNSHIP • TASK SUBMISSION • EXECUTIVE SUMMARY

Transaction records	250,000	Unique customers	49,661
Analysis period	01 Jan 2020 – 13 Sep 2023	Supplied churn label	20.0%
Repeat-customer share	96.7%	Inactive >180 days	51.1%

EXECUTIVE SUMMARY

This project analyzes customer transaction behavior to identify purchase patterns, build RFM-based customer segments, review repeat-purchase retention and propose engagement actions for Alfido Tech. The primary file, **ecommerce_customer_data_large.csv**, was analyzed separately from the second CSV in the supplied ZIP; the files were not merged.

METHOD

Data cleaning and validation were followed by customer-level Recency, Frequency and Monetary (RFM) feature engineering. Three K-means clusters were built using log-transformed and standardized RFM features. Category revenue, monthly transaction volume, customer inactivity and monthly cohort retention were examined.

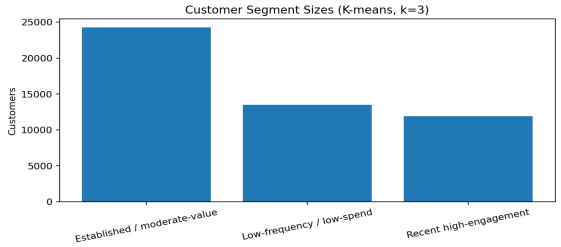
KEY FINDINGS

The dataset contains a supplied customer-level Churn label (20.0%). Separately, 51.1% of customers had no recorded purchase in the 180 days before the dataset's final date. This inactivity group is a practical re-engagement audience, not confirmed churn. The customer segments and purchase categories are profiled in the detailed report.

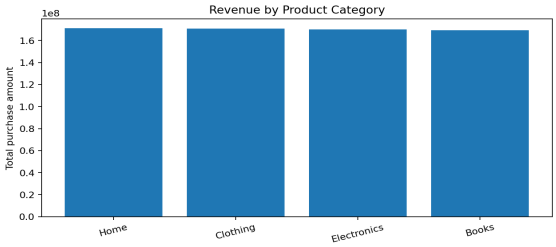
FIVE ACTIONABLE RECOMMENDATIONS

1) Run tailored win-back journeys for customers inactive for 181+ days. 2) Encourage second purchases with post-purchase reminders and complementary-product suggestions. 3) Reward recent high-value customers through loyalty benefits and early access. 4) Use category-specific bundles and cross-sells rather than blanket discounts. 5) Improve return-status completeness and monitor cohort retention, repeat rate and reactivation monthly.

VISUAL SNAPSHOTS



Customer segments



Revenue by category

CODE & DELIVERABLES

GitHub repository / notebook code: https://github.com/Punith137/Alfido-Customer-Behavior-Analysis
Included: Jupyter notebook, detailed PDF report, this one-page summary, README and two visualizations.

LIMITATIONS

Each row is treated as a transaction because no separate transaction ID is supplied. Missing Returns values remain unknown (19.0% of rows). Currency is unspecified. The churn-label definition is undocumented; inactivity is a separate proxy and K-means is descriptive, not a predictive churn model.